CHAPTER 7

Numbers

What is a computer? The answer to that question today would be something along the lines of "an electronic device that stores, retrieves, and manipulates data." In the seventeenth century, the answer would have been "a person that computes" (said person would be engaged in performing calculations, usually with a mechanical aid such as an abacus). The original computers were humans that were good with numbers.

A computer programmer today, especially in the context of web development, is probably more concerned with processing text than crunching numbers (although text processing is an abstraction of mathematical operations). PHP is not a natural choice for "doing math" with a computer: other languages exist that are primarily designed for such tasks. A statistician would probably prefer to work in R, an engineer would build models in Matlab, and mathematicians might opt for Wolfram's Mathematica or the venerable (and highly performant) FORTRAN. Python is currently a popular choice for data scientists, and rumor has it that Perl is still out there in the wild and helping bioinformaticians on a daily basis…

Nevertheless, PHP is built with C, which is a general-purpose language with a rich set of mathematical features, and PHP exposes much of that functionality. Indeed, PHP's built-in mathematics functions are often wrappers around C's standard <math.h> and <stdlib.h> libraries. PHP was originally conceived as a pre-processor for HTML; today it is a general-purpose language in its own right, with a rich set of features and functions for processing numeric data.

In this chapter, we will be covering the following:

- Number systems and formatting
- Language operators
- Type casting and coercion
- Built-in mathematical functions
- Limits

© Paul Tregoing 2025
P. Tregoing, *Exploring PHP 8*, https://doi.org/10.1007/979-8-8688-2136-3_7